Radxa ROCK 4B

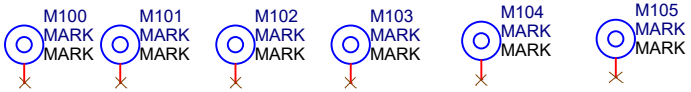
schematic

V1.52

CONTENT INDEXING

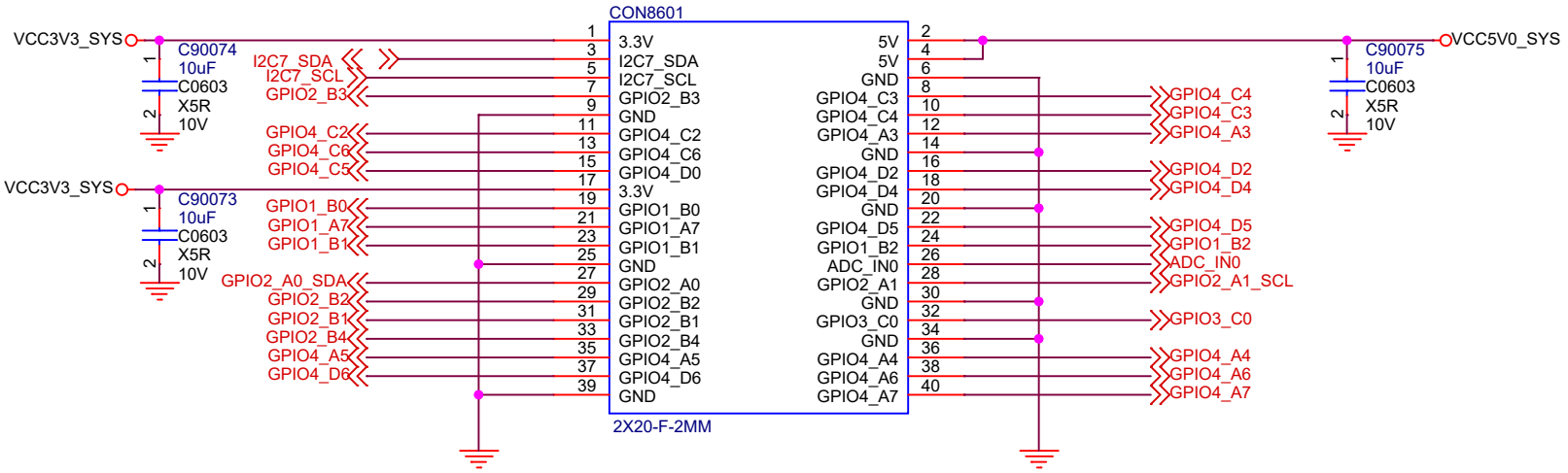
01.Cover
02.Index
03.Connector
04.ES8316
05.PCIe NGFF/M.2
06.RK3399 Power
07.RK3399 PMU Controler
08.RK3399 DDR Controler
09.RK3399 Flash&SDMMC Controler
10.RK3399 USB/USIC Controler
11.RK3399 SARADC/Key
12.RK3399 DVP Interface
13.RK3399 Display Interface
14.RK3399 GPIO
15.RK3399 PCIE
16.Power-DC IN
17.Power-PMIC RK808-D
18.RTL8211F
19.USB3.0 OTG Port
20.RAM-LPDDR4(200P)
21.Memory-EMMC
22.WIFI/BT AP6255
23.TF Card
24.HDMI Output

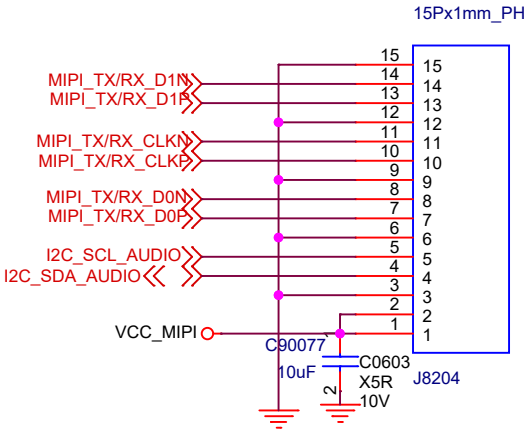


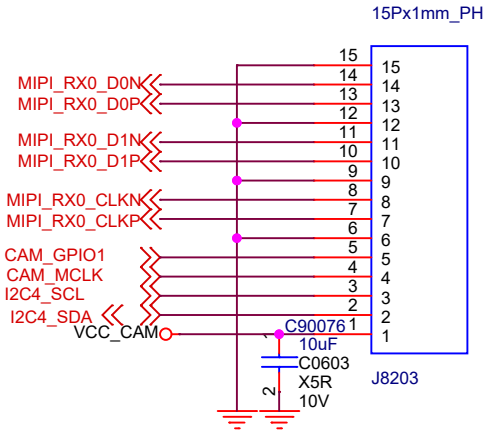


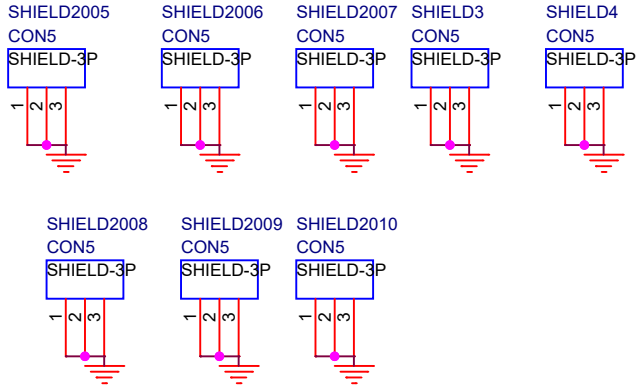
radxa

Size	Title:	REV
A4	ROCK 4B	
	Page Name: Index	1.52
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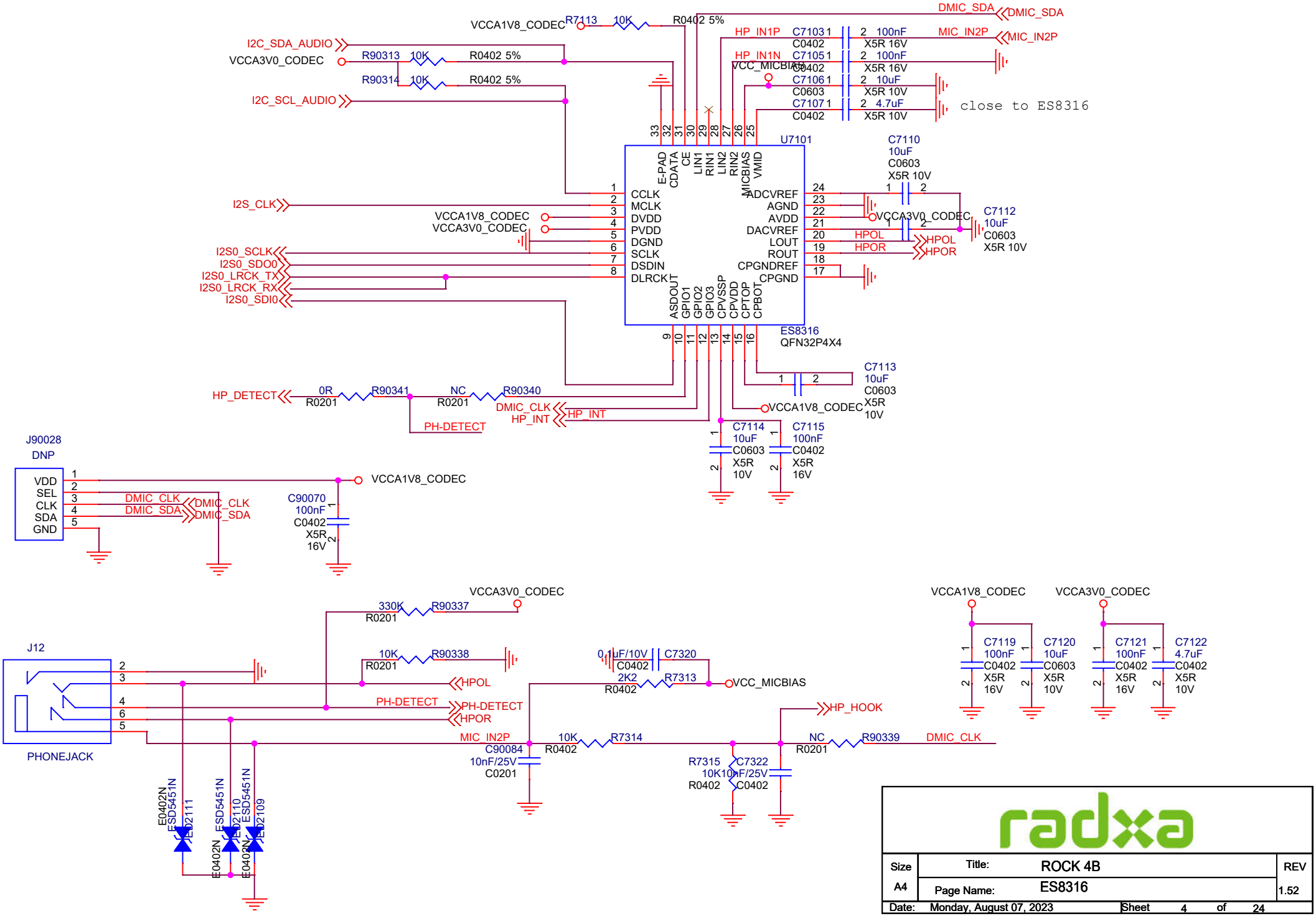




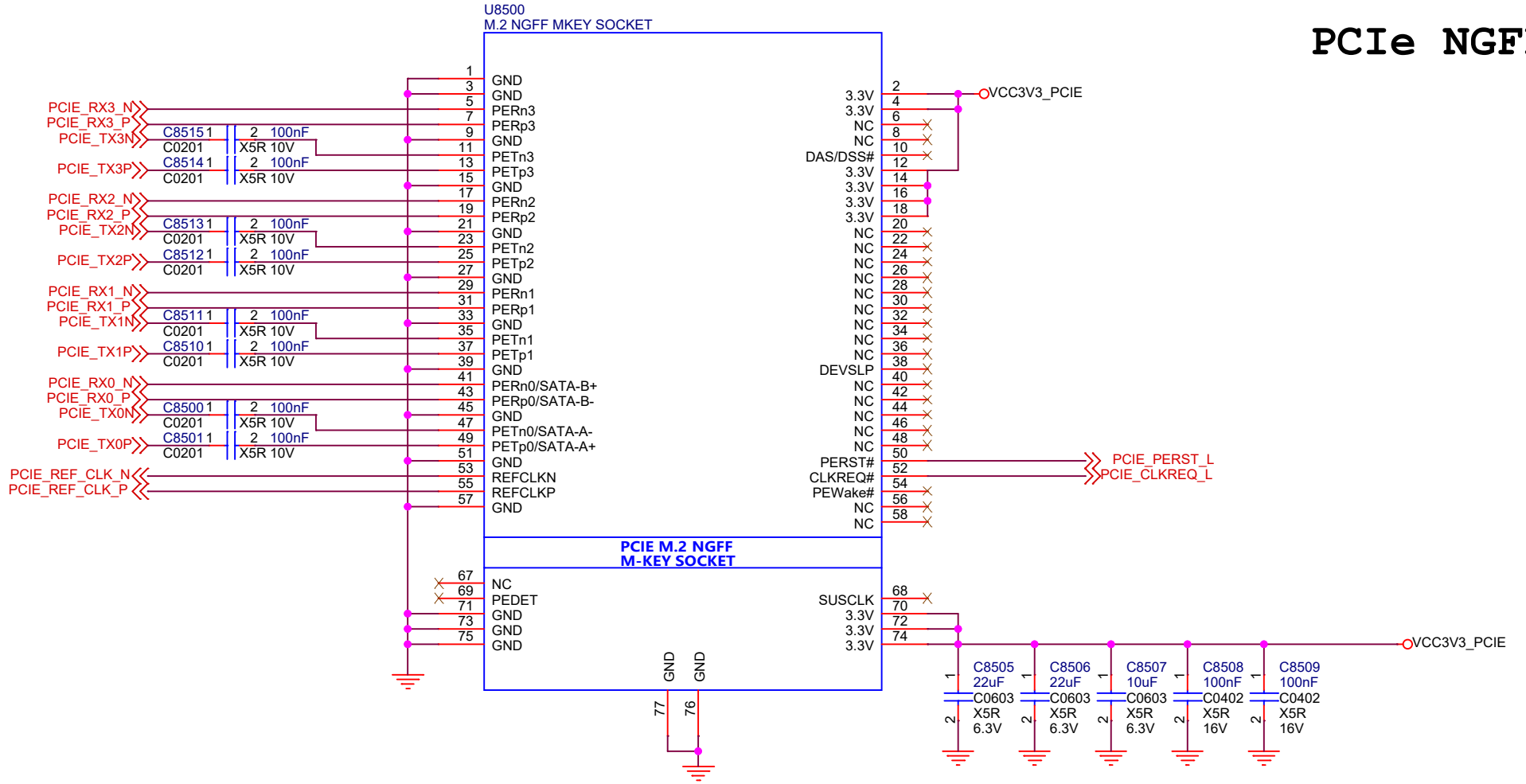


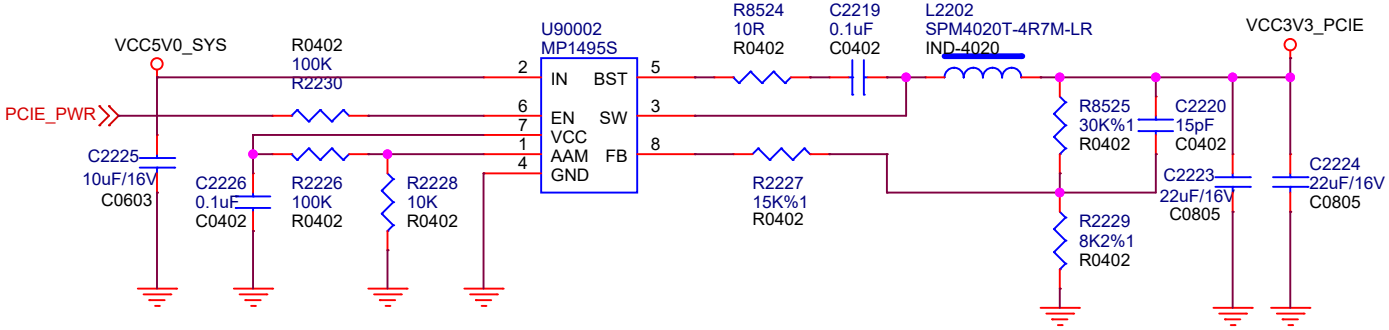


Size A4	Title: ROCK 4B	REV 1.52
	Page Name: CONNECTOR	
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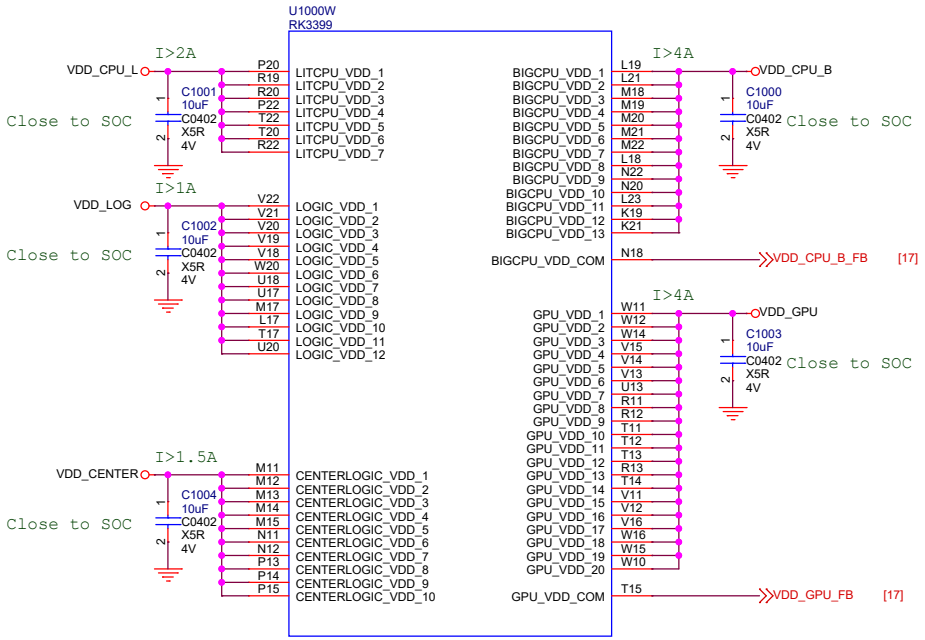

VCCA1V8_CODEC
R7113 10K R0402 5%
DMIC_SDA
I2C_SDA_AUDIO
VCCA3V0_CODEC
R90313 10K R0402 5%
R90314 10K R0402 5%
I2C_SCL_AUDIO
HP_IN1P C7103 1 2 100nF C0402 X5R 16V
MIC_IN2P
HP_IN1N C7105 1 2 100nF C0402 X5R 16V
VCC_MICBIAS
C7106 1 2 10uF C0603 X5R 10V
C7107 1 2 4.7uF C0402 X5R 10V
close to ES8316
U7101
E-PAD CDATA CE LIN1 RIN1 LIN2 RIN2 MICBIAS VMID
CCLK MCLK DVDD PVDD DGND SCLK DSDIN DLRCK
ADCVREF AGND AVDD DACVREF LOUT ROUT CPGNDREF CPGND
ASDOUT GPIO1 GPIO2 GPIO3 CPVSSP CPVDD CPTOP CPBOT
ES8316
QFN32P4X4
C7110 10uF C0603 X5R 10V
C7112 10uF C0603 X5R 10V
I2S_CLK
I2S0_SCLK
I2S0_SDO0
I2S0_LRCK_TX
I2S0_LRCK_RX
I2S0_SDI0
HPOL
HPOR
HP_DETECT
0R R0201 R90341
NC R0201 R90340
PH-DETECT
DMIC_CLK
HP_INT
C7113 10uF C0603 X5R 10V
C7114 10uF C0603 X5R 10V
C7115 100nF C0402 X5R 16V
J90028
DNP
VDD SEL CLK SDA GND
C90070 100nF C0402 X5R 16V
J12
PHONEJACK
330K R0201 R90337
10K R0201 R90338
0.1uF/10V C7320 C0402
2K2 R0402 R7313
VCC_MICBIAS
HP_HOOK
MIC_IN2P
C90084 10nF/25V C0201
10K R0402 R7314
NC R0201 R90339
R7315 10K R0402
C7322 10nF/25V C0402
E0402N ESD5451N LED2111
E0402N ESD5451N LED2110
E0402N ESD5451N LED2109
C7119 100nF C0402 X5R 16V
C7120 10uF C0603 X5R 10V
C7121 100nF C0402 X5R 16V
C7122 4.7uF C0402 X5R 10V
radxa
Size A4
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PCIe NGFF/M.2

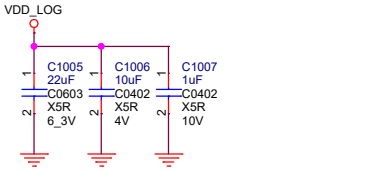


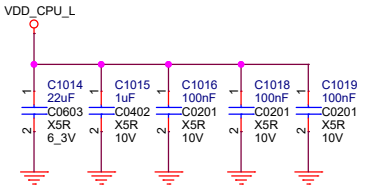


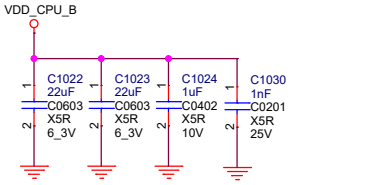
Size A4	Title: ROCK 4B		REV 1.52
	Page Name: PCIe NGFF/M.2		
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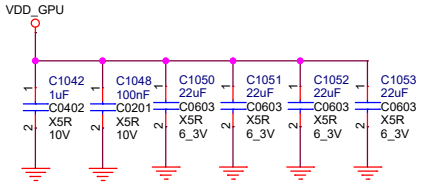


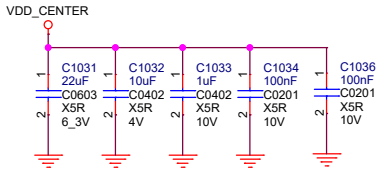
Note:Power filter CAP please place back of SOC or close to SOC

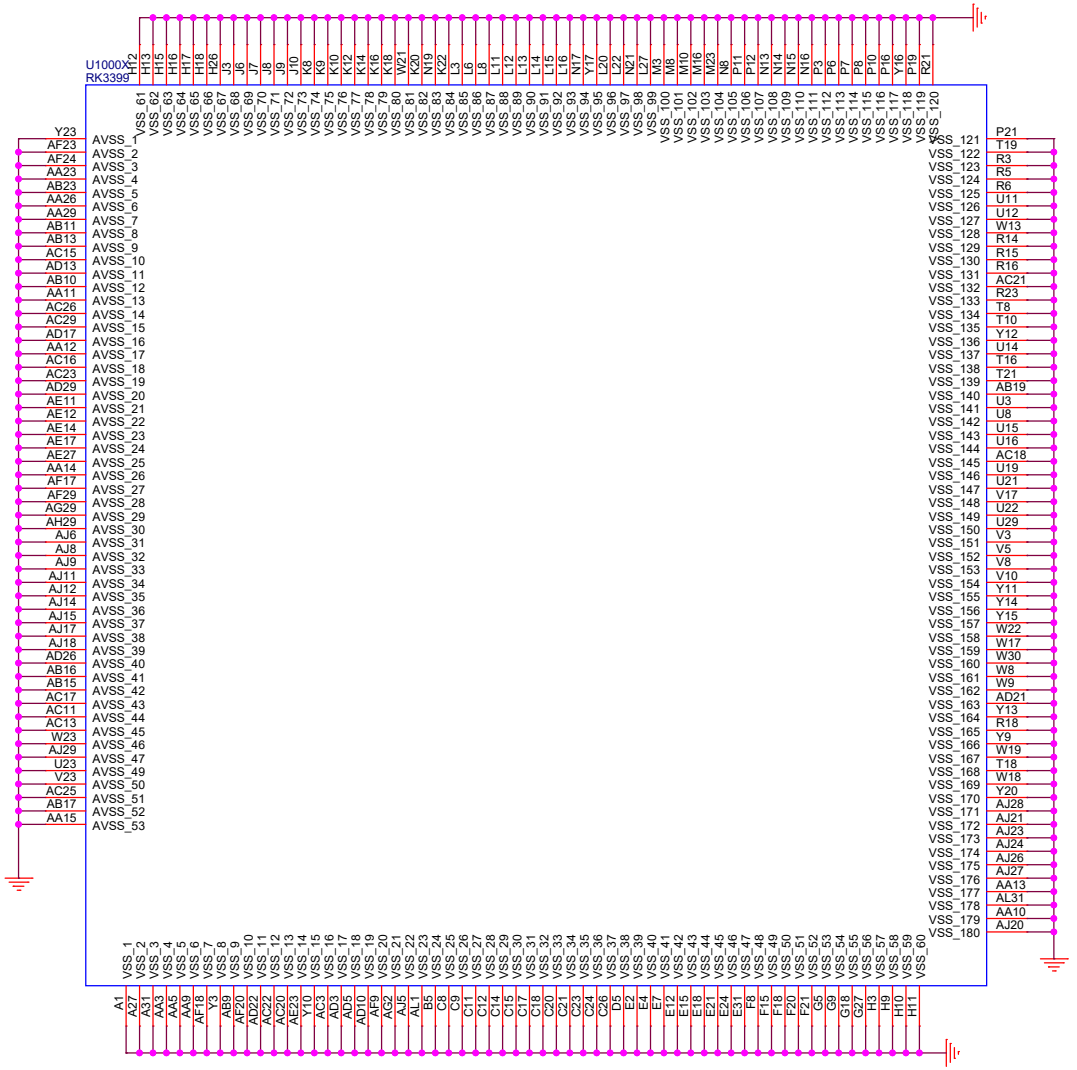




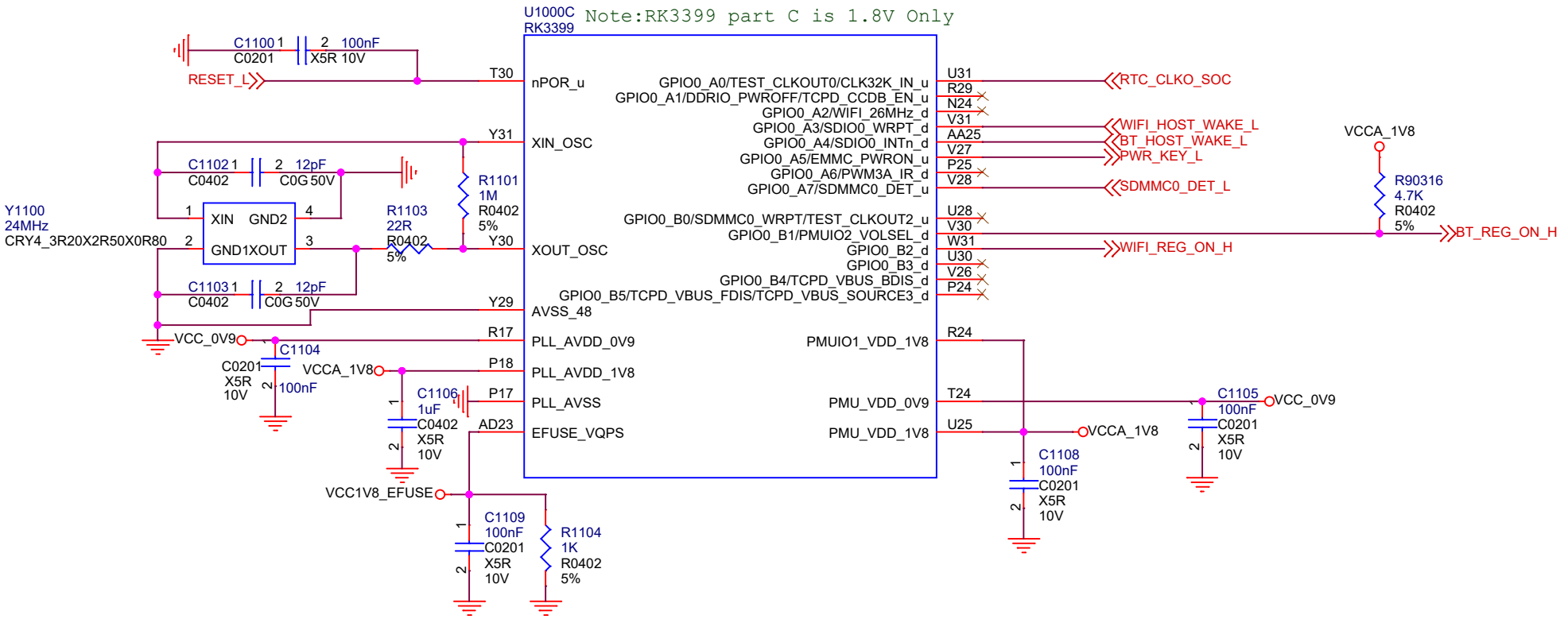




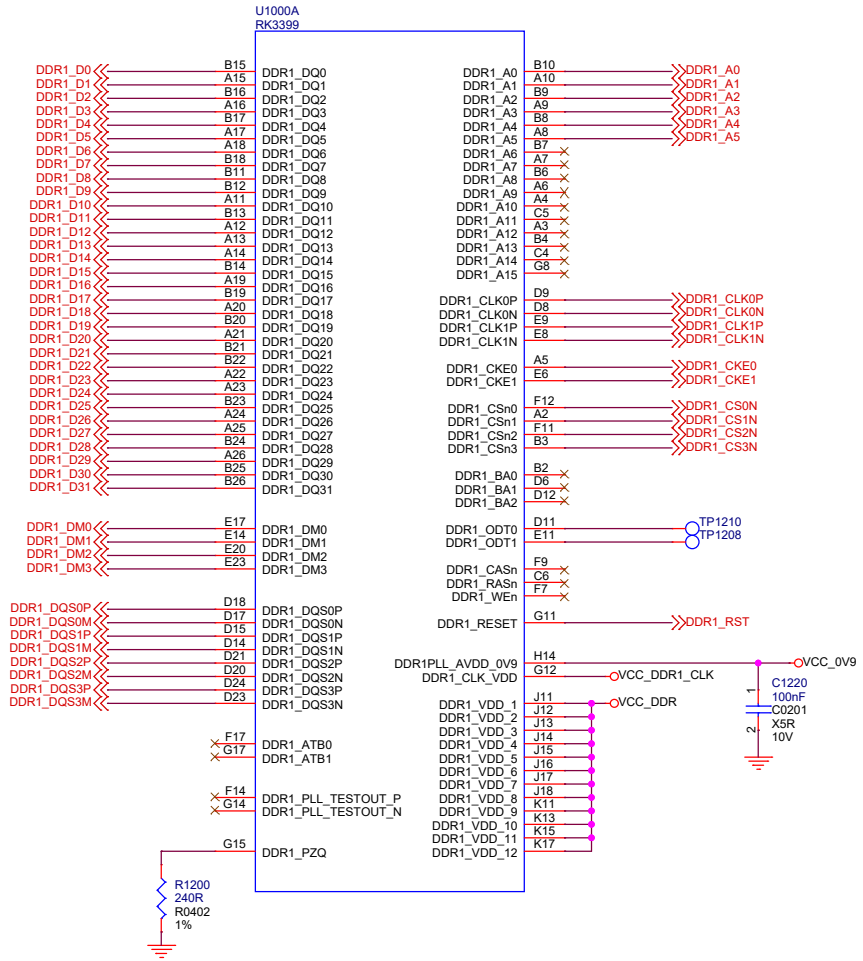


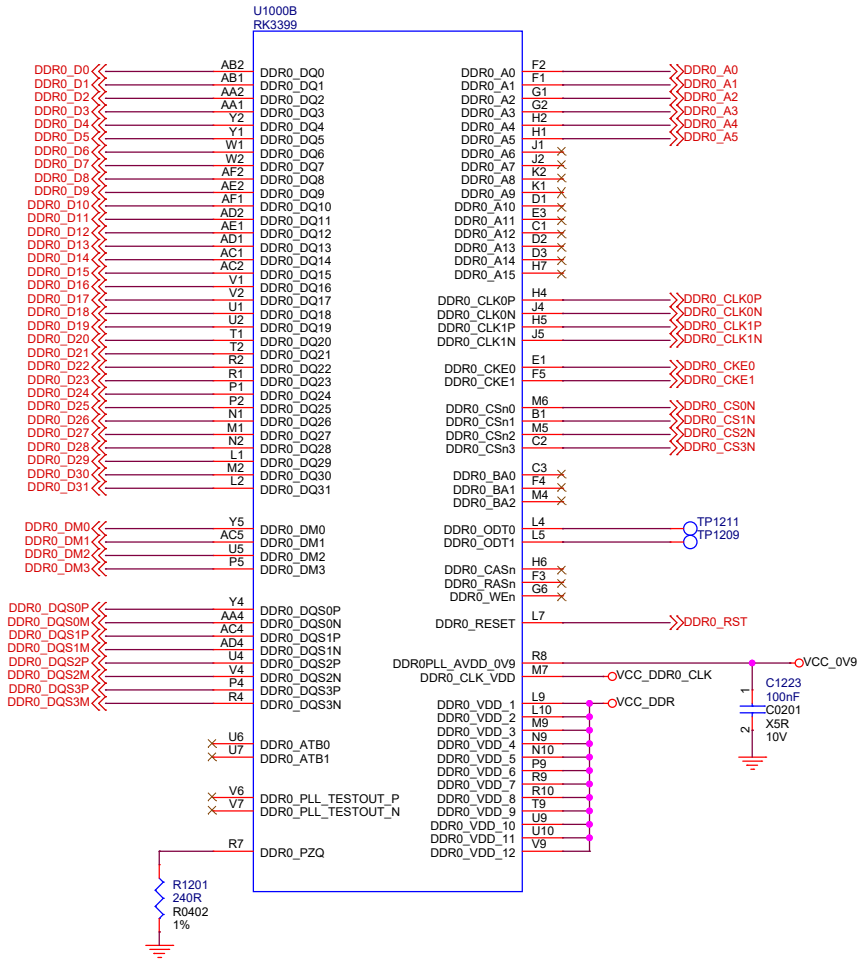


Size: A3	Title: ROCK 4B		REV: 1.52
	Page Name: RK3399 Power		
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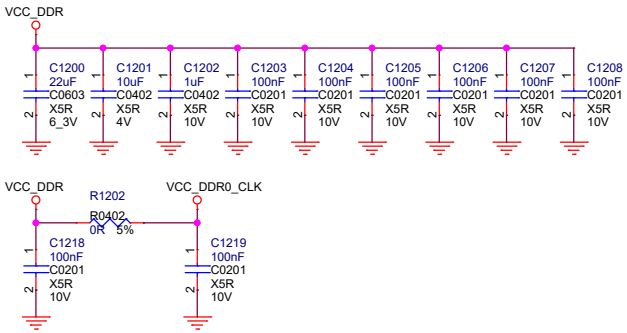


Size A4	Title: ROCK 4B	REV 1.52
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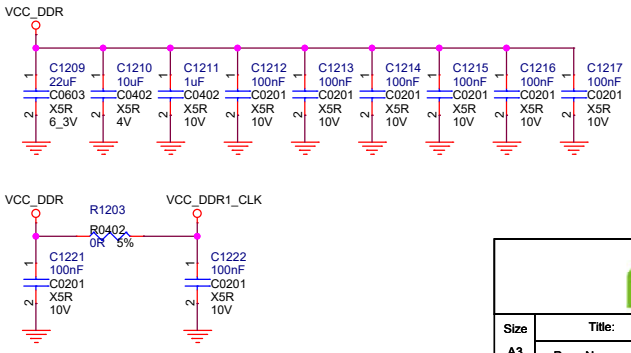




DDR FILTER Note:R1202 cannot be deleted

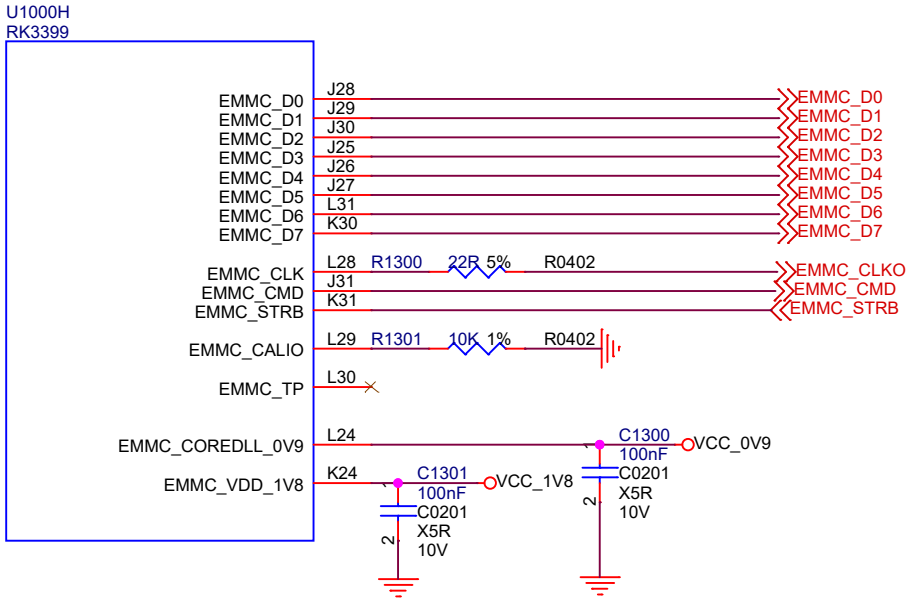


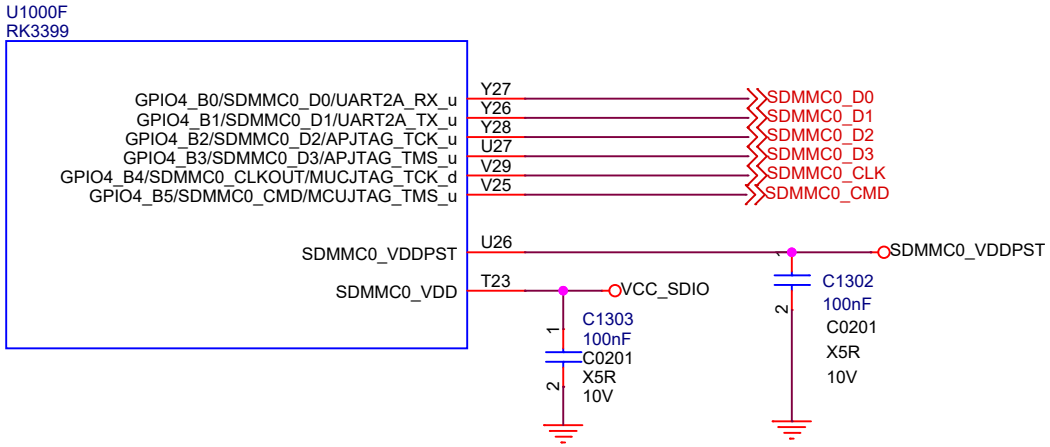
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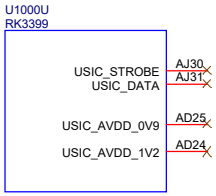
radxa

Size	Title:	ROCK 4B	REV
A3	Page Name:	RK3399 DDR Controler	1.52
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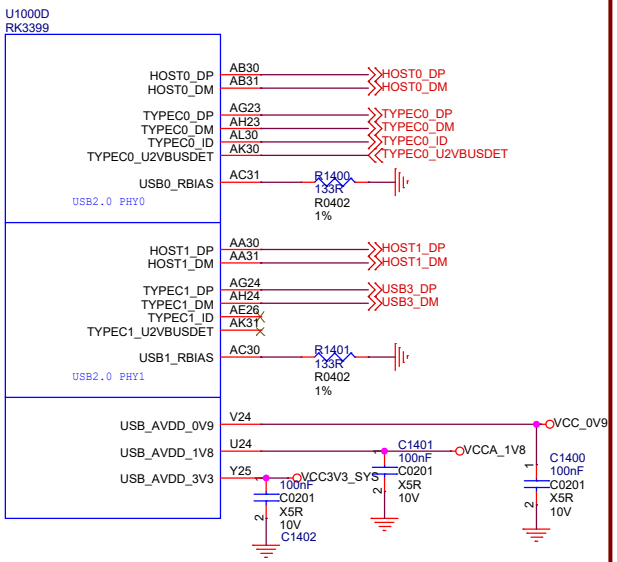




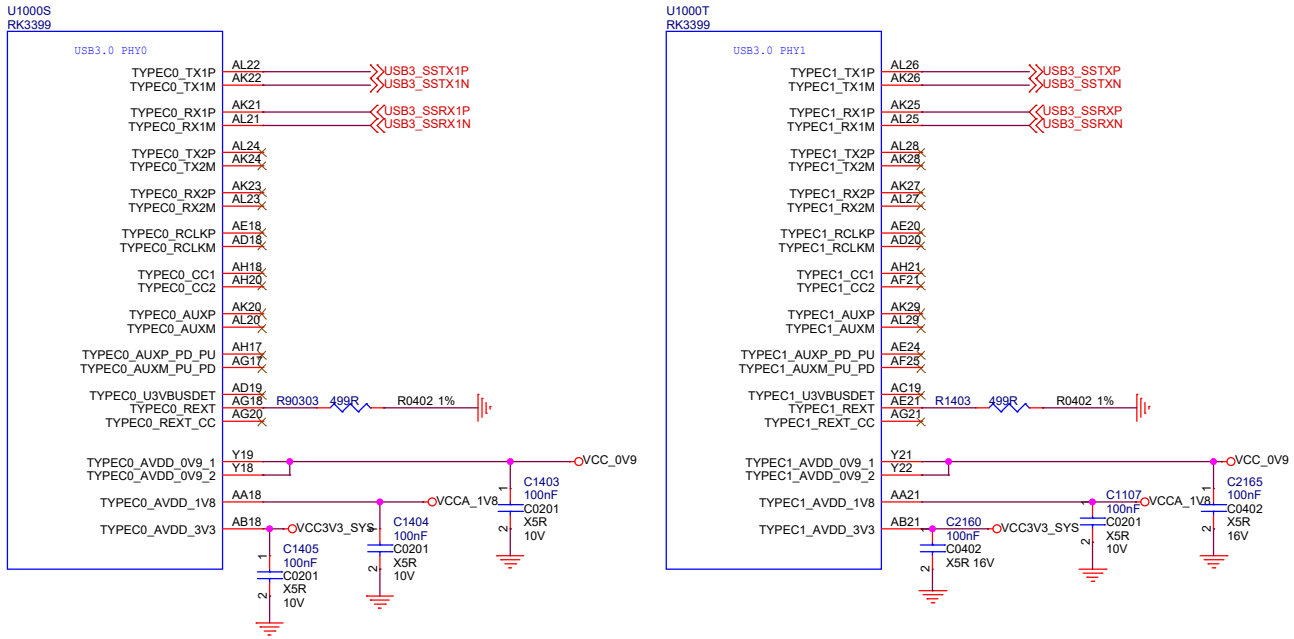
Size A4	Title: ROCK 4B	REV 1.52
	Page Name: RK3399 Flash&SDMMC Controler	
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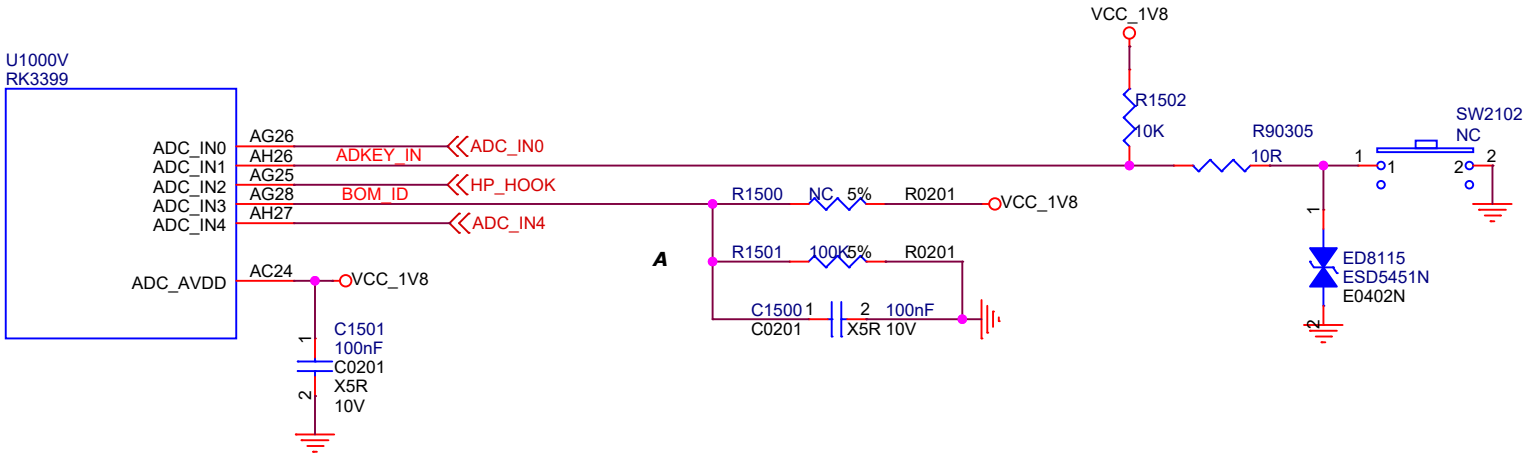
USB2.0



USB3.0



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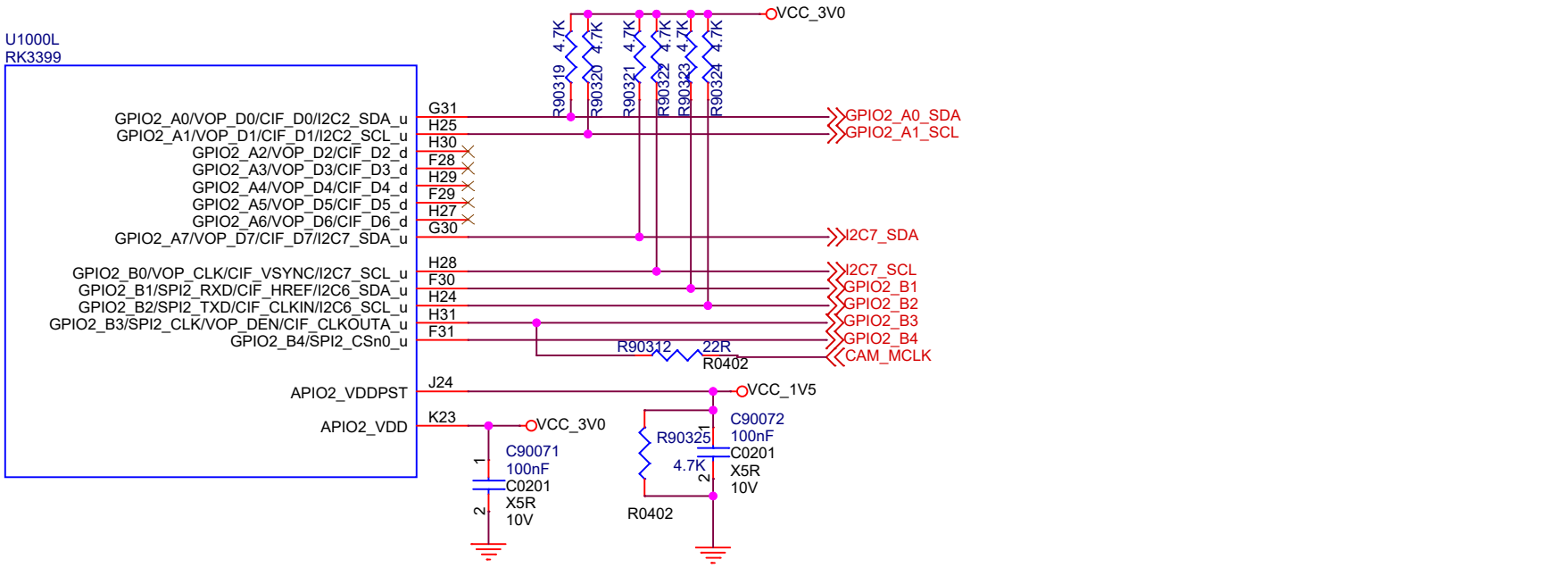
KEY BAORD

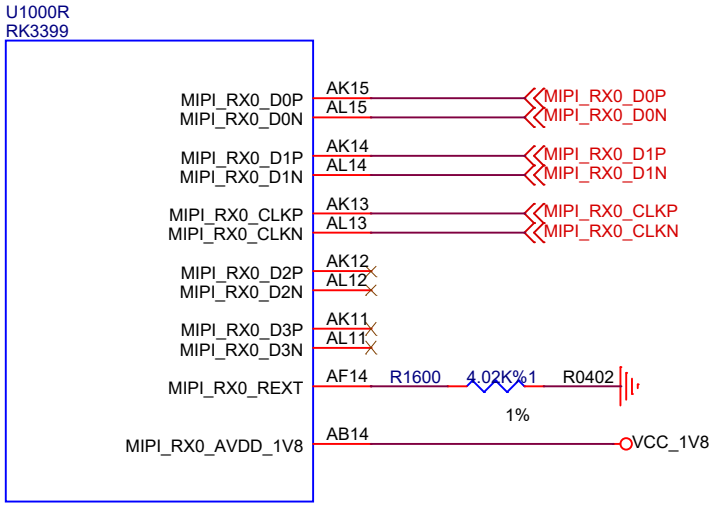
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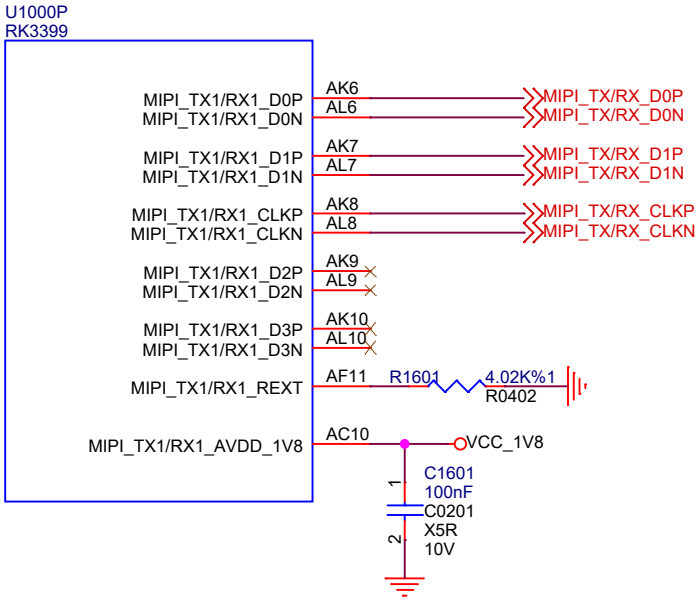
Item	Rup	Rdown		VOL	VERSION
LEVEL1	DNP	100K		0V	A
LEVEL2	100K	20K		0.3V	B
LEVEL3	100K	51K		0.6V	C
LEVEL4	100K	100K		0.9V	D
LEVEL5	100K	200K		1.2V	E
LEVEL6	100K	499K		1.5V	F
LEVEL7	100K	DNP		1.8V	G

radxa

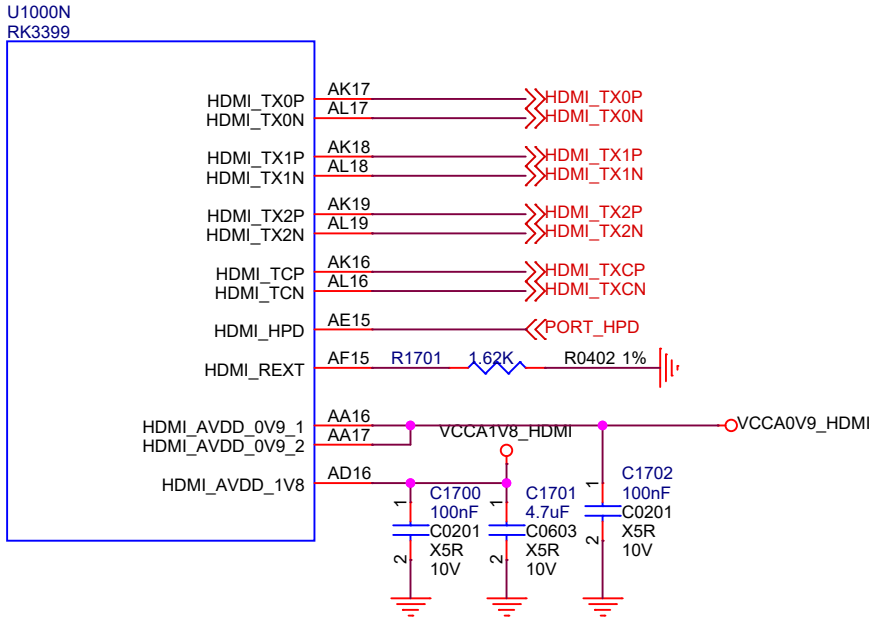
Size	Title:	ROCK 4B	REV
A4	Page Name:	RK3399 SARADC/Key	1.52
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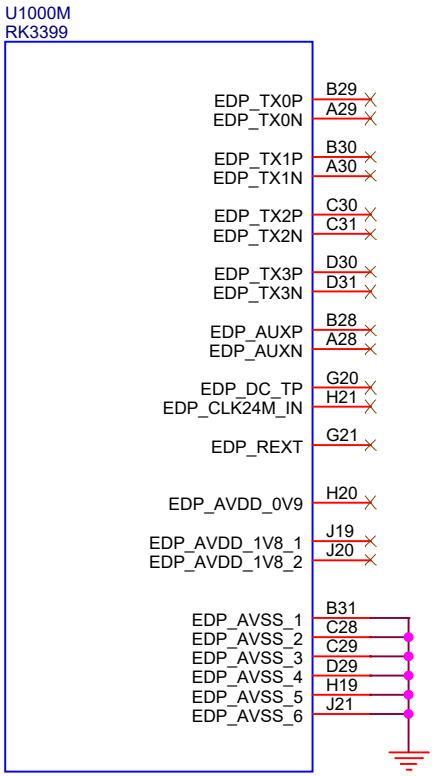


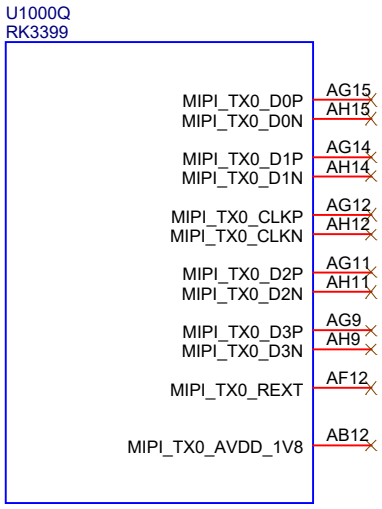




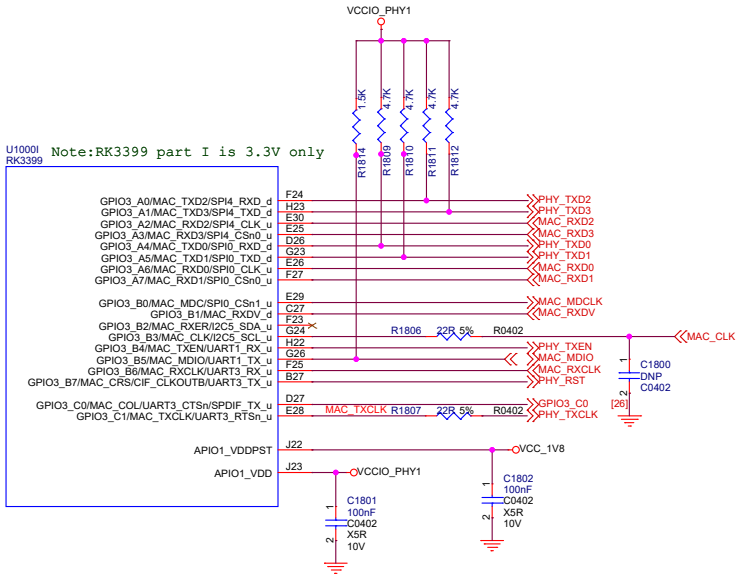
Size	Title:	REV
A4	ROCK 4B	
	Page Name: RK3399 DVP Interface	1.52
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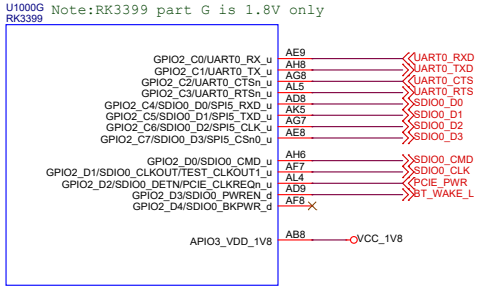


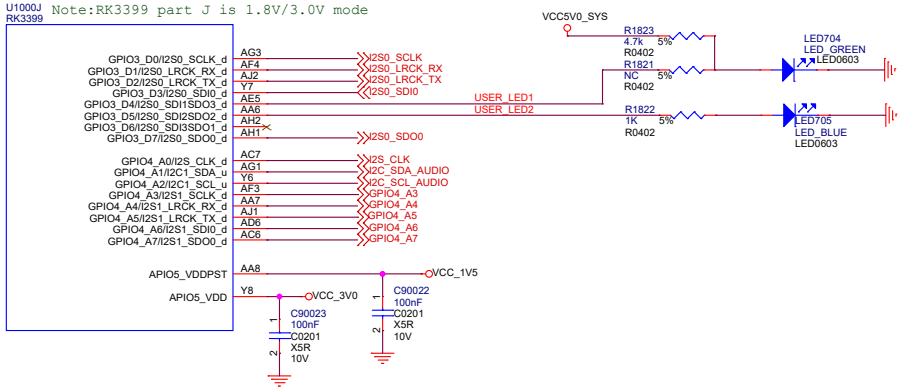


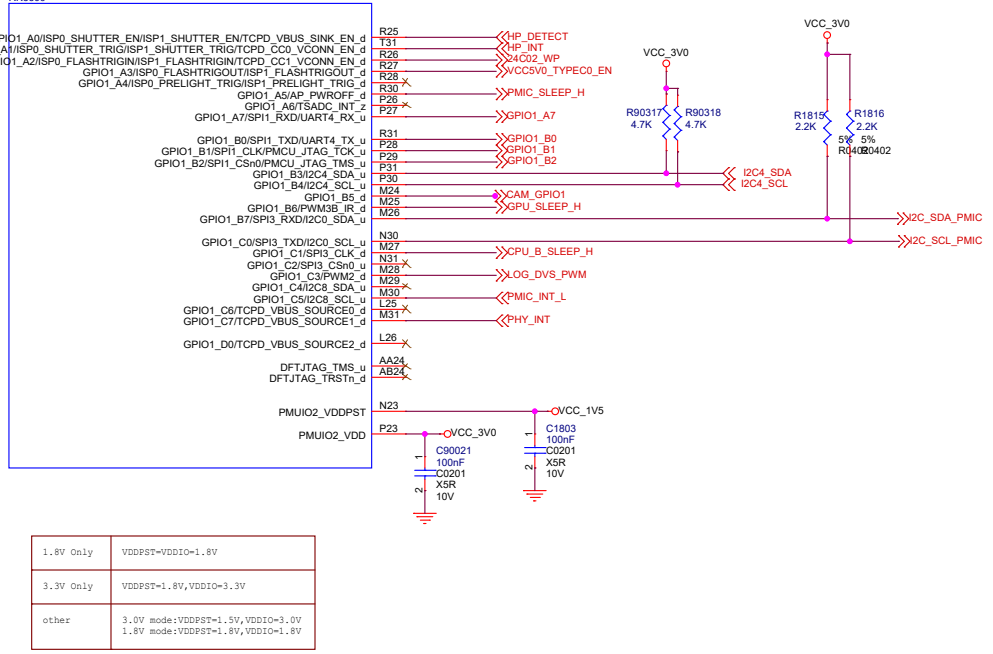


Size: A4	Title: ROCK 4B Page Name: RK3399 Display Interface	REV: 1.52
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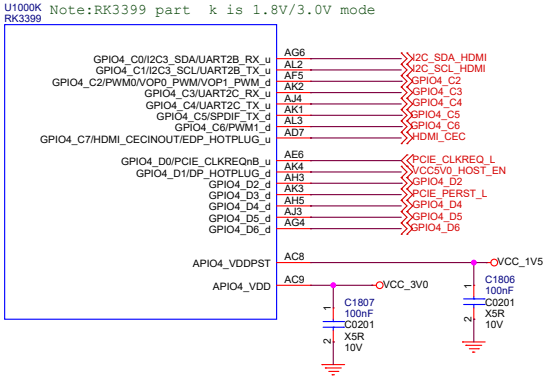






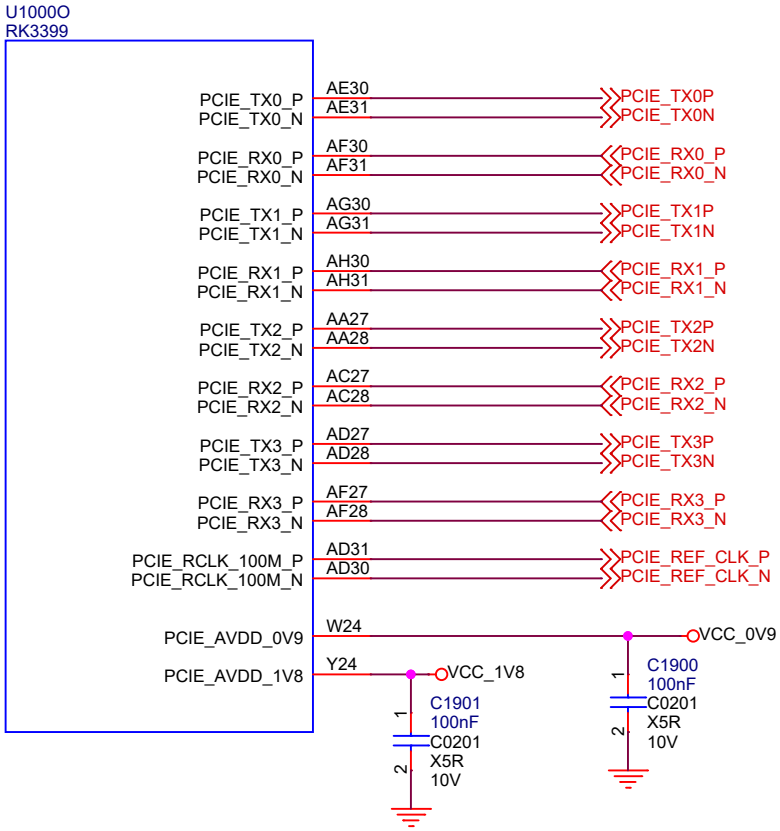


1.8V Only	VDDPST=VDDIO=1.8V
3.3V Only	VDDPST=1.8V,VDDIO=3.3V
other	3.0V mode:VDDPST=1.5V,VDDIO=3.0V 1.8V mode:VDDPST=1.8V,VDDIO=1.8V



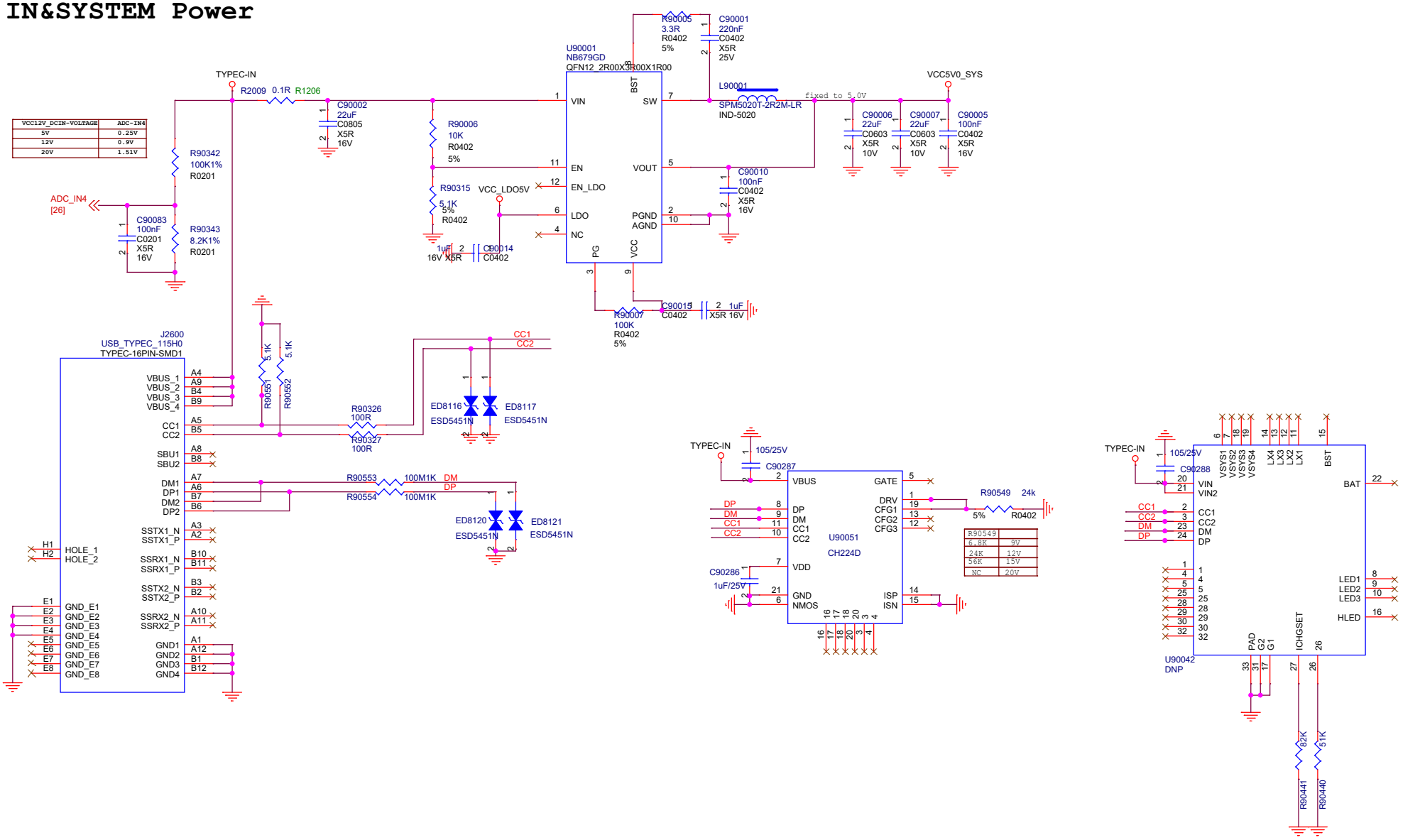
radxa

Size	Title: ROCK 4B	REV
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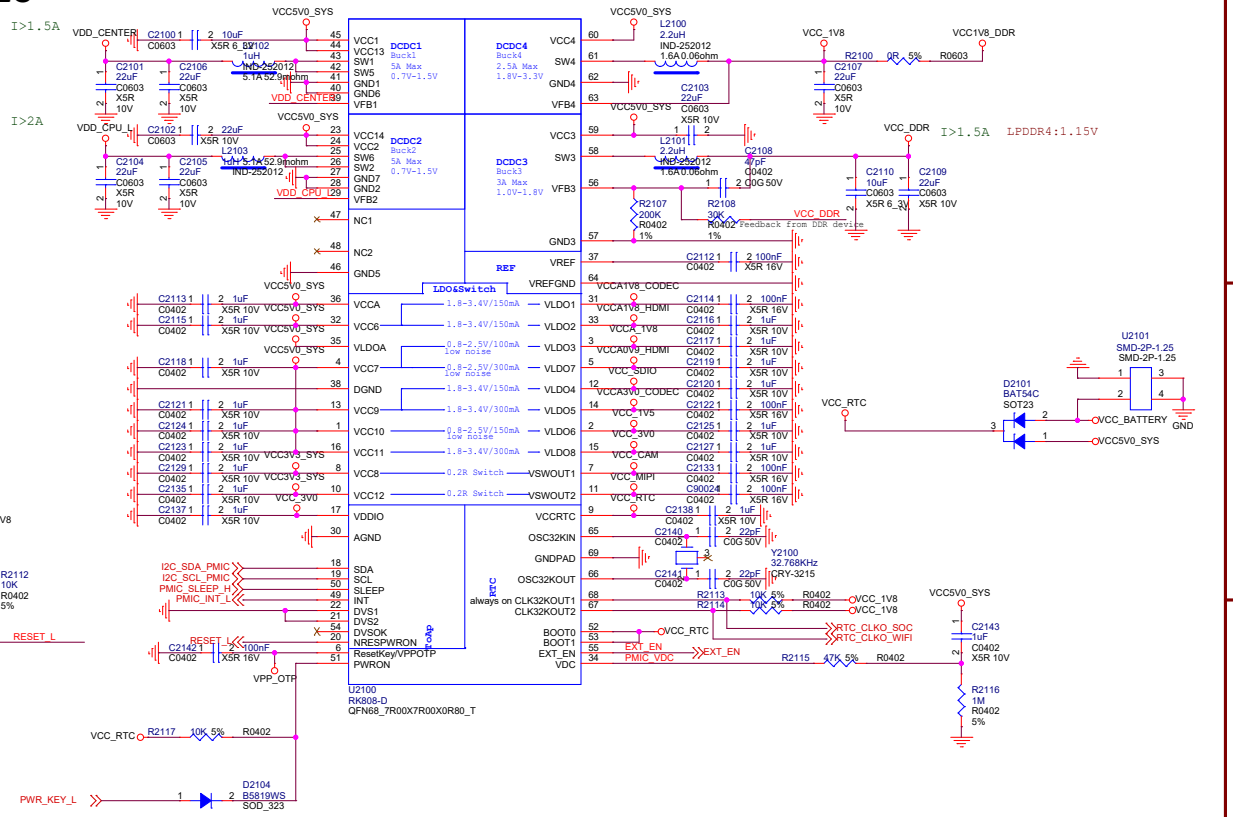
Size A4	Title: ROCK 4B	REV 1.52
	Page Name: RK3399 PCIE	
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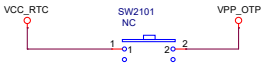
DC IN&SYSTEM Power



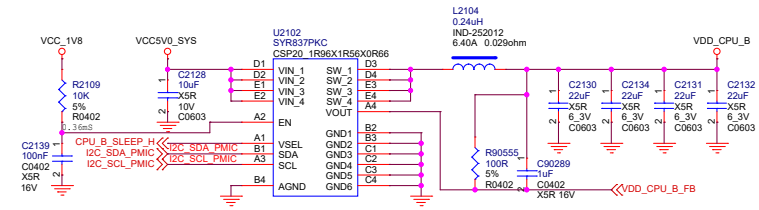
Size	Title:	ROCK 4B	REV
A3	Page Name:	Power-DC IN	1.52
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PMIC

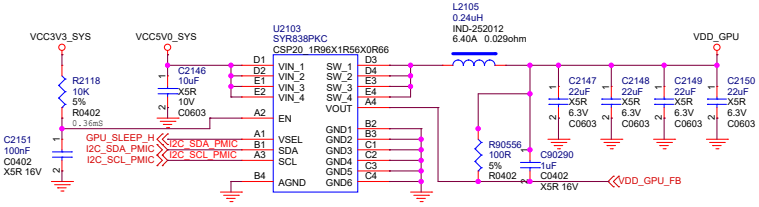




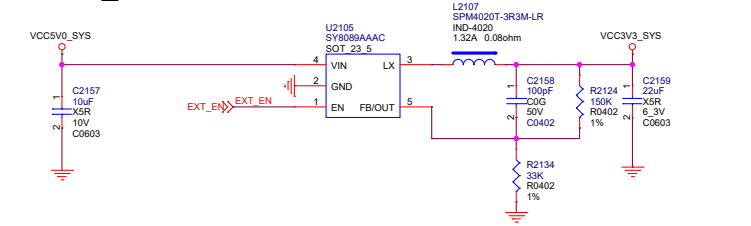
VDD_CPU_B power



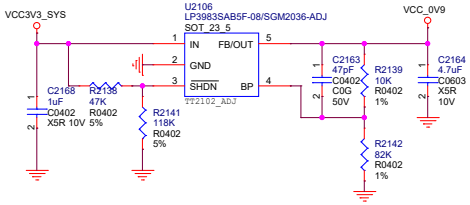
VDD_GPU power



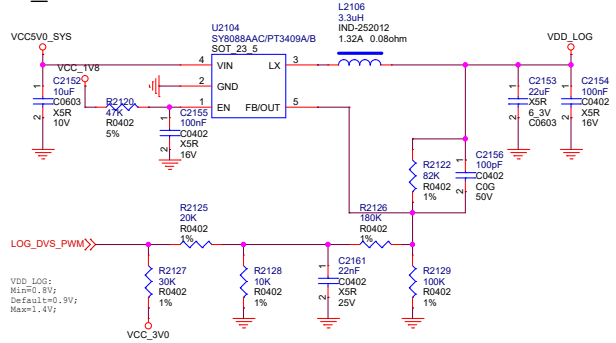
VCC3V3_SYS power

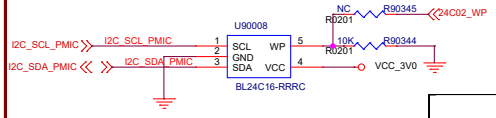


VCC_0V9 power



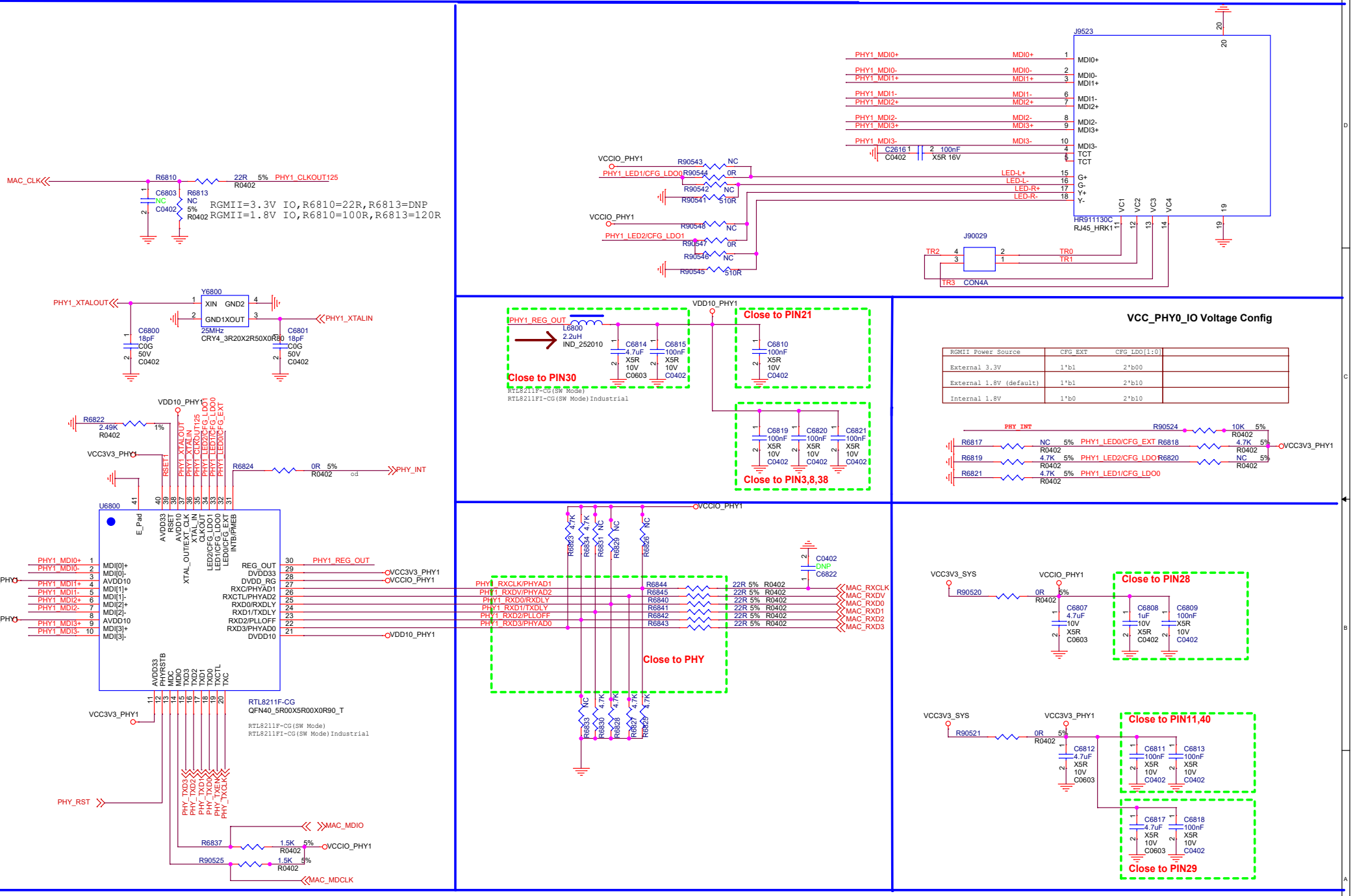
VDD_LOG power



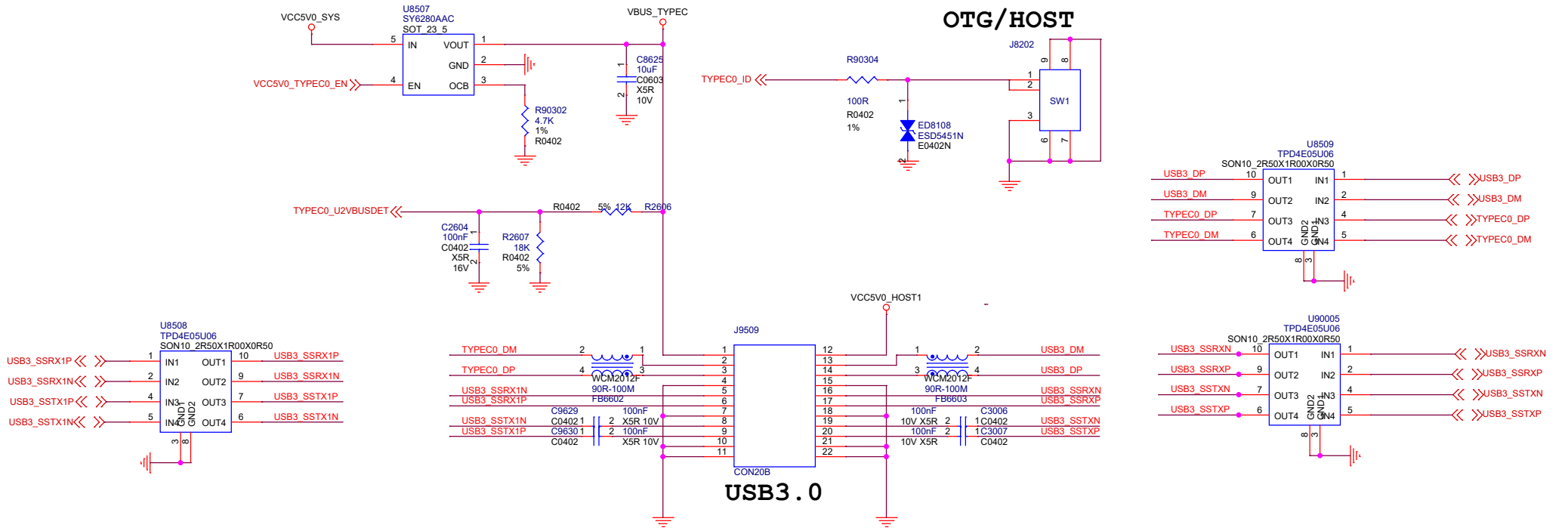


radxa

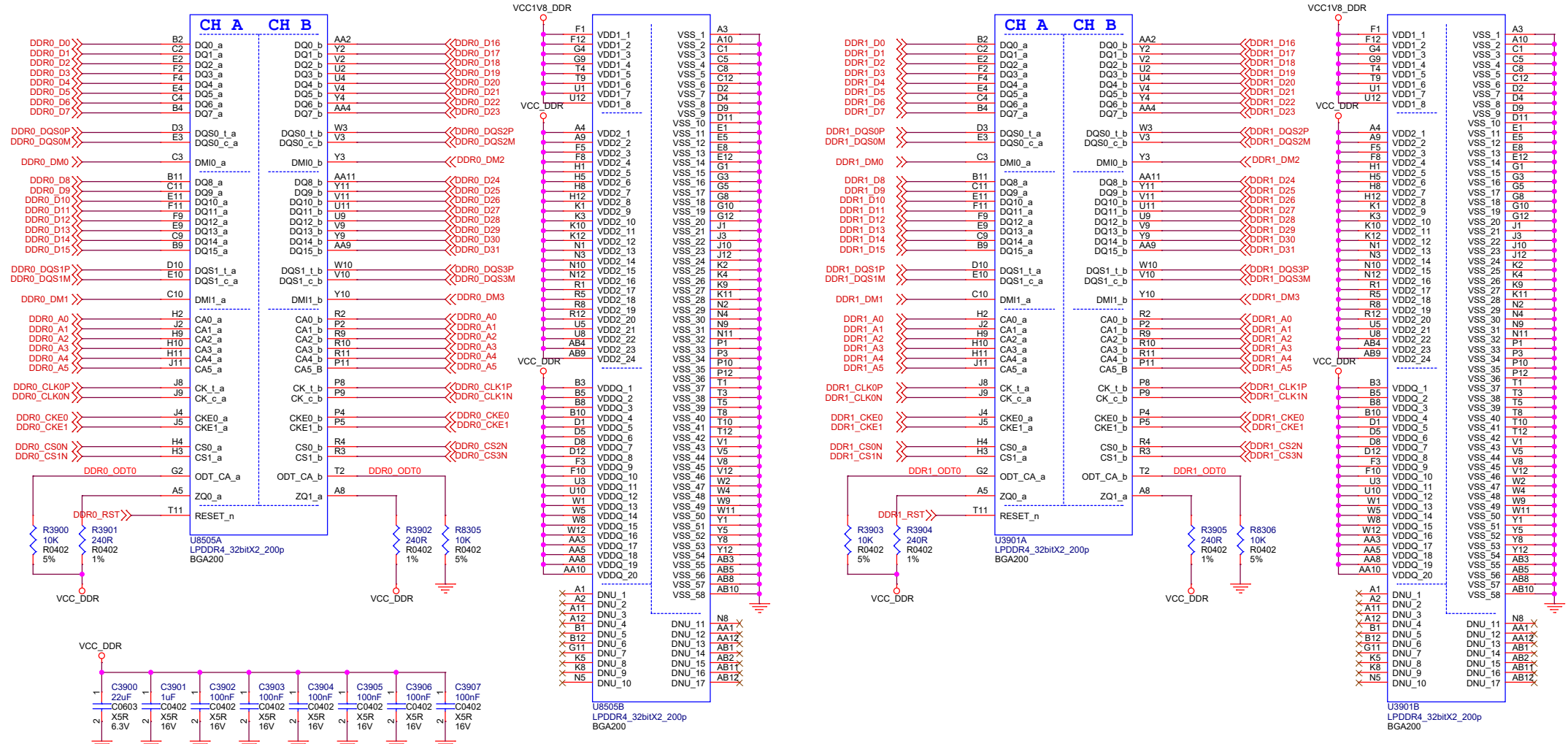
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Custom	Page Name: Power-PMIC RK808-D		1.52
Date: Monday, August 07, 2023	Sheet 17	of 24	

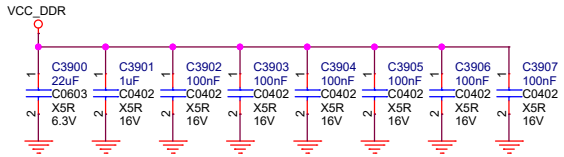


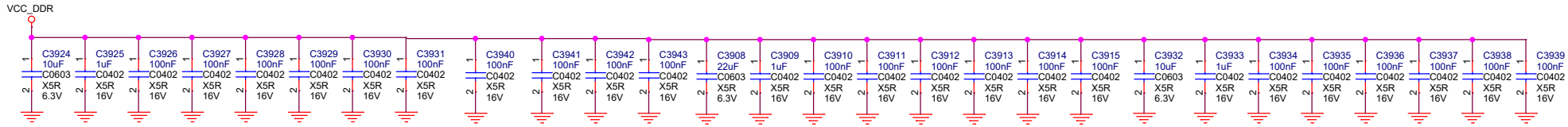
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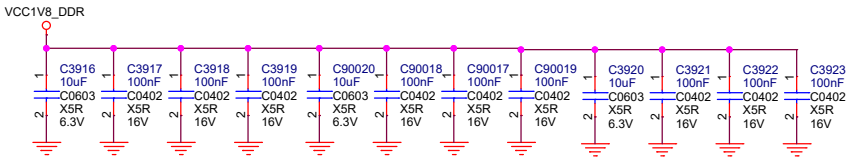


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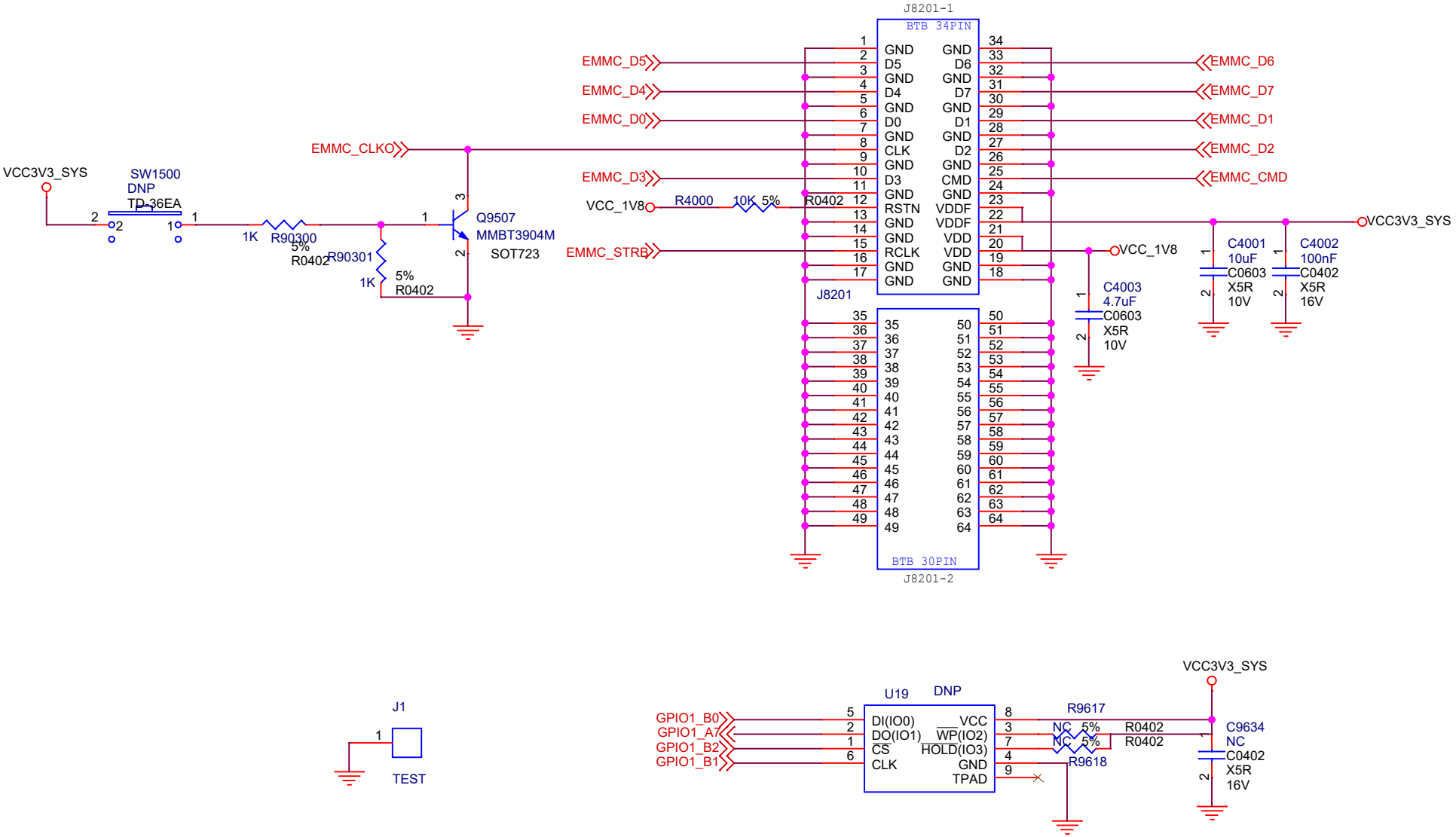




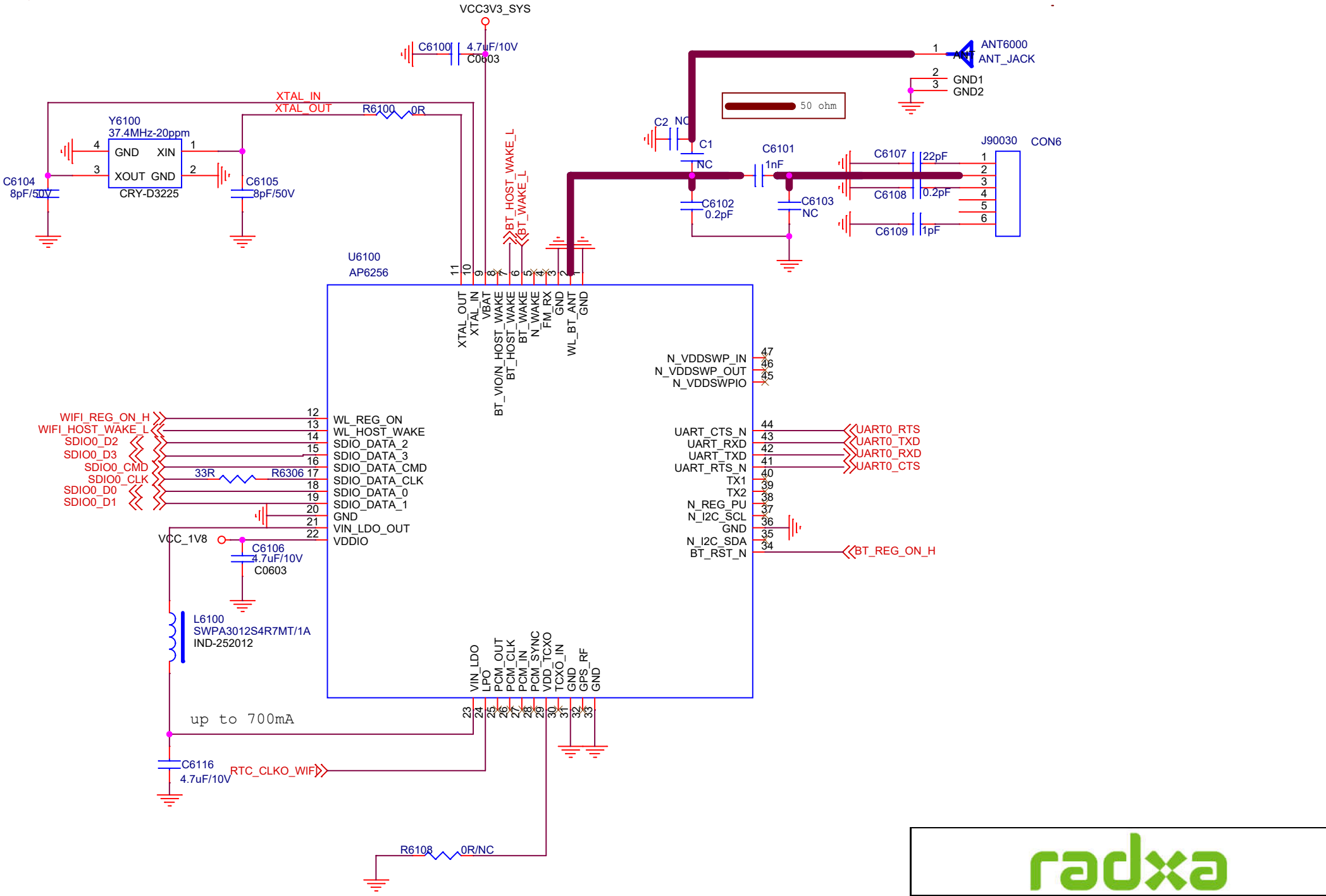
radxa

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eMMC

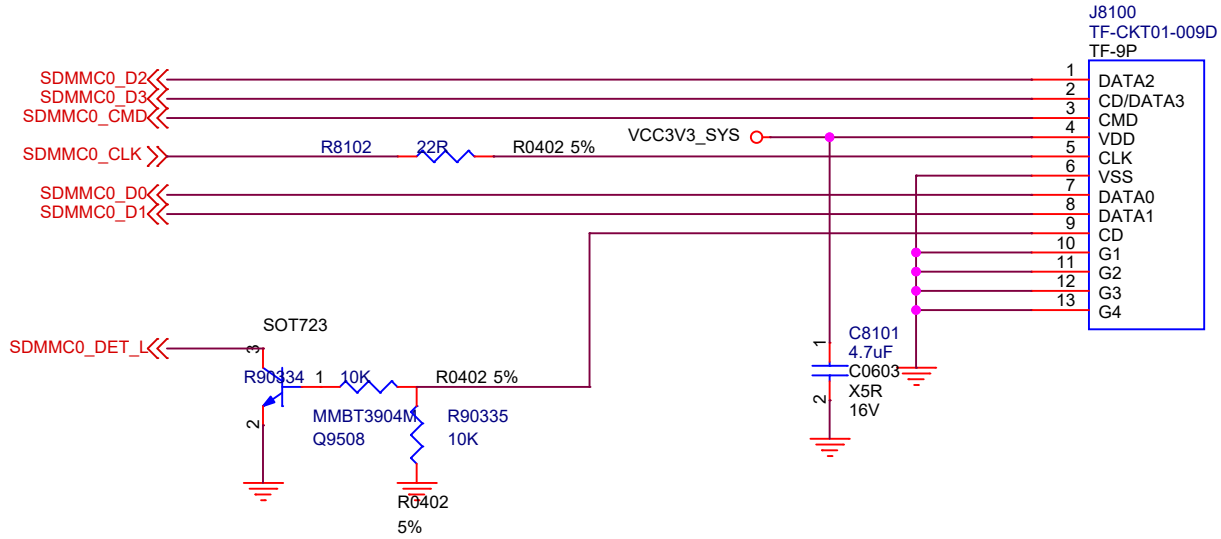


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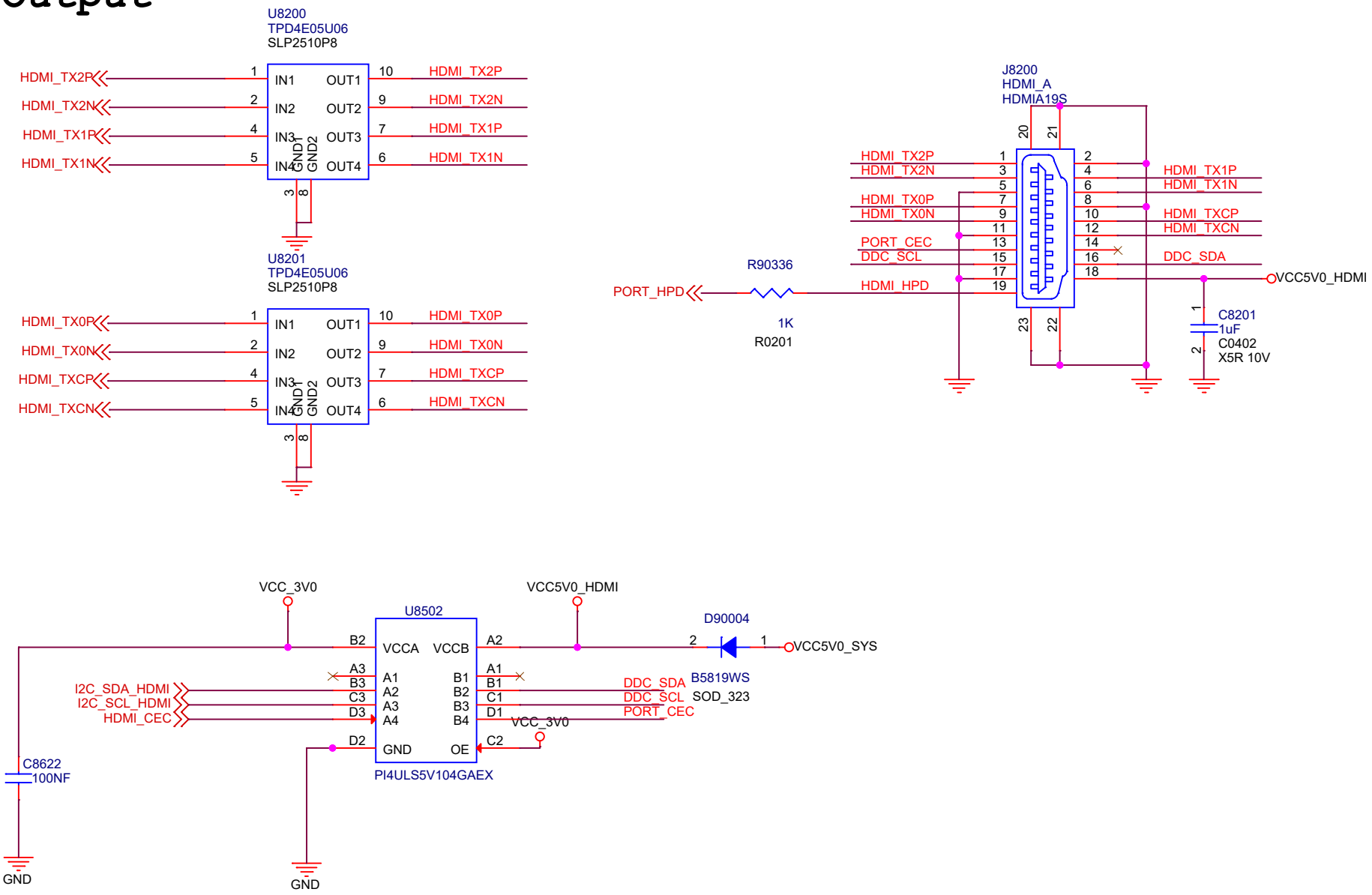
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TF CARD



Size: A4	Title: ROCK 4B	REV: 1.52
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HDMI Output



Size A4	Title: ROCK 4B		REV 1.52
	Page Name: HDMI Output		
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